1.1 INTRODUCTION TO BPCL-KR

1.1.1 OVERVIEW OF BPCL- KOCHI REFINERY

BPCL Kochi Refinery, a Strategic Business Unit (SBU) of Bharat Petroleum Corporation Ltd (BPCL), has a crude oil refining capacity of 15.5 MMTPA and its major products include LPG (Liquefied petroleum gas), Motor Spirit, Naphtha, Aviation Turbine Fuel, Kerosene, Special Boiling Point Spirit (SBPS), Mineral Turpentine Oil (MTO), High Speed Diesel, Light diesel oil, Furnace oil, LSHS, bitumen, Natural Rubber Modified Bitumen, aromatic products like Benzene & Toluene and petrochemical like propylene. BPCL Kochi Refinery is the largest PSU refinery in India.

BPCL-Kochi Refinery is located at Ambalamugal in Ernakulum District in Kerala State. The company was incorporated in 1963 as Cochin Refineries Ltd, a joint venture company of Government of India, Phillips Petroleum USA, and Duncan Brothers. In 1969 Duncan Brothers divested their shares and in November 1988, Phillips Petroleum Corporation divested its entire holding.

This Refinery which had an initial refining capacity of 2.5 MMTPA was dedicated to the nation on 23rd September 1966. Cochin Refineries Ltd was then renamed as Kochi Refineries Ltd (KRL) in May 2000. Govt. of India had divested its entire holding in favor of Bharat Petroleum Corporation Ltd (BPCL). Pursuant to Order dated 18 August, 2006 issued by Ministry of Company Affairs Kochi Refinery Ltd. has been amalgamated with Bharat Petroleum Corporation Ltd.

From the date of commissioning to till date, the refinery undertook major expansions like addition of refining capacity, setting up secondary processing units like FCCU, Aromatics Recovery Unit, fuel gas desulphurization unit, Diesel Hydro De-Sulphurization unit, Capacity Expansion & Modernization Projects etc. The company implemented Integrated Refinery Expansion Project (IREP) on 2017 which escalated the company’s crude processing capacity from current 9.5 MMTPA to 15.5 MMTPA with additional secondary and tertiary processing facilities. BPCL-Kochi Refinery became the largest PSU refinery in India.

1.1.2 GROWTH PATH:

The company's expansion alleyway chronologically given below:

1966: 2.5 MMTPA Crude Unit -I

1973: Capacity Expansion from 2.5 MMTPA to 3.3 MMTPA

1984: Capacity Expansion from 3.3 to 4.5 MMTPA & 1.0 MMTPA of FCCU

1989: Aromatic Recovery Unit (87200 TPA Benzene & 18000 TPA Toluene)

1991: Captive Power Plant (21 MW)

1993: Direct marketing of Benzene & Toluene

1993: Poly Isobutene Unit (5200 TPA Joint venture with Balmer Laurie & Co.)

1994: 3 MMTPA Crude Unit II, Fuel Gas SRU & FCCU revamp (1.4 MMTPA)

1995: Steam Turbine Generator (17.8 MW)

2000: 2.0 MMTPA DHDS and SRU

2003: LPG bottling plant of 44,000TPA

2004: Bitumen emulsion plant 10,000 TPA

2005: DHDS Revamp to meet the BSII specifications for auto fuels and capacity expansion of DHDS to 2.536 MMTPA.

2005: FCCU Capacity expansion to 1.75 MMTPA

2007: SINGLE POINT MOORING (SPM) facilities at Puthu Vypeen was commissioned in Nov'2007 and the first ship berthed at SPM on 3rd Dec'2007

2007: Biturox unit for producing Bitumen. (3, 80,225 TPA)

2008: Propylene Recovery Unit (50,000 TPA)

2009: Capacity expansion of CDU-II to 5.0 MMTPA

2010: GT2 (34.5MW) commissioned

2010: NHT/CCR (0.85 MMTPA) commissioned

2011: VGOHDS (1.7 MMTPA)/ SRU (80 TPD) commissioned

2013: IREP Foundation Stone Laying (Jan 2013)

2013: Fuel conversion GT from Naphtha to RLNG

2013: Feed Conversion of HGU from Naphtha to RLNG

2017: Became the largest PSU refinery in India. IREP units CDU 3(10.5 MMTPA), DHDT (4.3 MMTPA), VGOHDT (3.0 MMTPA), SRU-3, DCU (3.84 MMTPA), PFCCU (2.2 MMTPA), NHT-ISOM (0.36 MMTPA), are commissioned.

2019: The petrochemical project of BPCL Kochi Refinery viz. Propylene Derivatives Petrochemical Project (PDPP) envisages utilization of 250,000 MT Propylene available from PFCCU for the production of niche or specialty derivatives viz. Acrylic Acid (160 KTPA), Oxo Alcohol (212 KTPA) and Acrylates (190 KTPA).

2020: MSBP to be commissioned to fulfill the commitment towards EURO-VI emission standards which will be applicable in PAN India basis from April 1, 2020 onwards. MSBP consists of following units: NHT (1.5 MMTPA), CCR (0.8 MMTPA), CYCLEMAX (908 kg/hr), PENEX (0.7 MMTPA), Hot oil system, Utilities- Cooling water system (6400 m^3/hr), Instrument air compressor (3500 Nm^3/hr).

1.1.3 PRODUCT BASKET:

All products of Kochi refinery are marketed by marketing SBU's of our corporation namely, Retail, LPG, Aviation and I&C. The products from Kochi Refinery are catering to a predominantly South Indian market by tanker trucks, Rail wagon, pipe lines and costal tankers. Further, being a coastal refinery, Kochi refinery exports products like Naphtha, Furnace oil as and when need arises.

1.1.4 CAPACITY & COMPLEXITY:

BPCL-Kochi Refinery (BPCL-KR) is processing 15.5 MMTPA of crude oil (imported and indigenous).Crude oil processing involves, Primary Distillation, Secondary Processing and Quality Up gradation Process. LPG, Naphtha, Kerosene, Diesel, VGO and Vacuum Residue are the products obtained during primary distillation including vacuum distillation. Naphtha is further processed in CCR & ARU to get Benzene, Toluene, SBPS & blend stream for Motor Spirit. Kerosene is further treated for ATF and MTO. Diesel is further processed for reducing sulfur content to meet BS VI specification. Vacuum residue is further treated / blended with other products for making Bitumen, Furnace oil and LSHS. VGO from primary distillation is further processed in FCCU, PFCCU& VGO HDS, VGO HDT yielding value added products like LPG, MS (Petrol), heavy Naphtha, LCO and CLO are produced.

1.1.5 AUXILIARY FACILITY:

The Refinery has an offsite facility with 8 crude tanks, 150 products, intermediate and 5 Horton spheres, 11 mounted bullets. The product and crude movement are through SBM/SPM & Cochin Oil Terminal (COT) for receiving crude oil from the tankers, North Tanker Berth and South Tanker Berth are mainly meant for product tanker loadings/unloading. Effluent treatment plants for treating the effluents to the specified Minimal National Standards (MINAS).

Refinery is provided with facilities for generation and transmission of utilities like Steam, Power, Nitrogen, Compressed Air, De-Mineralized Water, Raw Water and Cooling Water. These utilities facilitate various unit operations performed in a refinery. There are 11 steam boilers in addition to 3 Waste Heat Recovery Boilers and Process Steam Generators.

The combined electrical power requirement of BPCL-KR is around 110MW under full load operation of all processing plants. At CPP, electrical power is generated using two gas turbine generators – GTG1 (22MW) and GTG2 (34.5MW) and one steam turbine generator STG (17.8MW). At IREP CPP, GTG3 (25MW), GTG4 (20MW), GTG5 (20MW) are providing power. BPCL-KR also receives power from KSEB’s 220kV Ambalamugal substation, via two 33kV cable feeders – Grid Inc-1 & 2.

1.2 INTRODUCTION TO MOTOR SPIRIT BLOCK PROJECT (MSBP):

The project involves construction and commissioning of three new units:

- Naphtha Hydro treating Unit
- CCR Platforming Unit
- Penex Isomerization Unit

The project has been implemented in order to ensure compliance of MS specifications in accordance with the new BS-VI standards. The stricter sulphur specification requires additional reforming and isomerization units for ensuring maximum utilization of refinery naphtha. The units have been designed with the latest UOP technology and will lower operating expenses by reducing load on existing MS contributing units. The project is executed in LSTK mode through M/s Petrofac (Balance of units), M/s GR Engineering(CCR reactor and Cyclemax) and M/s Technicas Reunidas (Heaters).

1.2.1 Stream Hours and Turndown Ratio:

The units are designed for an on-stream factor (operating hours) of 8000 hours per annum towards arriving at the hourly processing rate and the units are designed for a minimum operating cycle of 3 years between turnarounds with units operating continuously at 100% capacity i.e. to process 3 x 365 = 1095 days design feed stock.

1.2.2. Plant Turndown Capacity:

The Units are designed for a turndown ratio (minimum operable hydraulic capacity) as mentioned below while making on-spec products.

• NHT Unit	:	50% of design capacity
• PENEX Unit	:	40% of design capacity
• CCR Platforming Unit	:	50% of design capacity

1.3 NAPHTHA HYDROTREATER UNIT

NHT unit is designed to process 1.5 MMTPA blend of straight run stabilized Naphtha (C5-125^{o}C), straight run heavy Naphtha (125^{o}C-140^{o}C), hydrotreated Naphtha and kerosene streams to reduce contaminants such as sulfur, Nitrogen, Olefin and metals. The hydrotreated and stripped Naphtha from the NHT unit is fractionated in a Naphtha Splitter column. The light Naphtha fraction is routed to Penex unit and heavy Naphtha fraction at the bottom is fed to CCR platforming unit. When kerosene is co-processed along with Naphtha, rerun column splits the heavy Naphtha at the top for CCR and hydrotreated kerosene at the bottom to storage.

1.4 CCR PLATFORMING UNIT

CCR Platforming Unit is designed to process 0.8 MMTPA of hydro treated and stripped heavy naphtha from the NHT splitter to produce a reformate product with a RONC of 103.0 min for gasoline blending. Net gas from the CCR Platforming Unit be processed in an integrated PSA System with tail gas recycle to produce 99.5 mole % min pure hydrogen. The Cyclemax-III is designed with a 907 kg/h (2000 lb/hr.) catalyst circulation for regenerating spent catalyst. Reduction Zone Vent Gas (RZVG) and Regeneration Vent Gas (RVG) Chlorsorb System is utilized to treat and recover chlorides from the combustion gases in CCR.

1.5 PENEX UNIT

Penex unit is designed to process 0.7 MMTPA of hydro treated and stripped Light naphtha from the NHT Unit. The Penex unit is designed with a Deisopentanizer (DIP) column & a Deisohexanizer (DIH) column to produce an isomerate product with RONC of 89.0 (min). Off gases from the penex unit will be processed in caustic wash scrubbing column and then dried in a net gas drying unit. The dry net gas will be sent to CCR platforming Unit to recover LPG and hydrogen.

1.6 HOT OIL SYSTEM

Apart from the NHT and CCR reactor feed heaters a large number of reboilers and intermediate feed heaters are required to meet the process parameters. To minimize the fired heaters and High-pressure steam consumption, a hot oil system is provided as a major heat input source. All heat required by the system is provided by a hot oil heater and convection coils of CCR heaters. The Hot oil used in the complex is Therminol 66, supplied by Eastman.

1.7 UTILITY SYSTEMS

1.7.1 Recirculating Cooling Water System

Dedicated Cooling tower (MUC-CT-101A/B/C) for MS Block is constructed by M/s Paharpur with 3 (2 Operating + 1 Standby) identical cells having capacity of 3200 m3/hr each. Cooling water is supplied to MS block using three (2 W +1S) cooling water circulation pumps, MUC-P-101A/B/C of capacity 3200 m3/hr each.

1.7.2 Instrument Air

Instrument air system is having 2 (1 Operating + 1 Standby) identical compressors of Ingersol Rand make with the capacity of 3500 Nm3/hr each to cater to the instrument air requirement of MSBP unit. Dedicated instrument and plant air distribution headers is provided which will supply the compressed air to all the consumers in MSBP.